

Atlas 210X — VFO Nano add-on wiring

YT3CT (ex YU5KBM) · Nano ATmega328P (Old Bootloader) · sketch atlas_vfo_fixed.ino · 2026-10 · button / relay part = proposal

1. LCD RG1602A (16×2) — series resistors + supply

Nano	Series R (at Nano end)	Signal	LCD pin
D7	470 Ω	RS	4
D8	470 Ω	E	6
D9	470 Ω	D4	11
D10	470 Ω	D5	12
D11	470 Ω	D6	13
D12	470 Ω	D7	14
GND	—	VSS, R/W	1, 5
+5 V	—	VDD	2
10k pot	wiper	V0 contrast	3
+5 V	(100 Ω on board)	Backlight A	15
GND	(0 Ω on board)	Backlight K	16

- Pins 7–10 (D0–D3) not connected.
- **At the LCD:** 100 nF ceramic + 10 μF between pin 2 (VDD) and pin 1 (VSS).
- Series R 220 Ω–1 k all OK (slows edges = less hash). Short cable along chassis, clip-on ferrite near the Nano.
- After fitting: adjust contrast pot (V0).

2. TTL programming leads (no USB needed)

USB-TTL cable (Prolific)	Nano
Green (TX)	D0 / RX
White (RX)	D1 / TX
Black (GND)	GND
Red (+5 V)	NOT connected
—	RST lead (+ optional push button to GND)

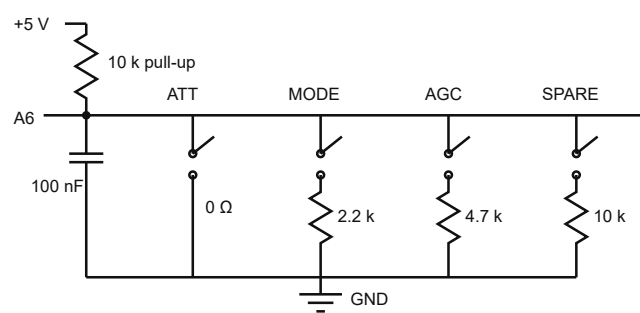
- If no sync: swap green / white (colours differ between cables).
- **Upload:** start upload, then short RST to GND for ½ s. Power cycling only works if the radio stays off until the Nano POW LED is fully out (TTL line back-feeds the Nano).
- Do the same on the VariBeam Nano (COM9 / via TTL COM20).
- Never plug the Nano USB in while the TTL cable is connected.

Free Nano pins & plan

Pin	Now	Plan
D0 / D1	Serial	TTL leads
D4	free	spare
D5	free	ATT relay driver
D13	free (LED)	AGC fast (2N7000)
A6	free (analog only)	button ladder
A7	free (analog only)	spare / 2nd ladder

Used: D2/D3 encoder, D6 enc. button, D7–D12 LCD, A0/A1 S-meter, A2 TX, A3 band, A4/A5 I²C Si5351.

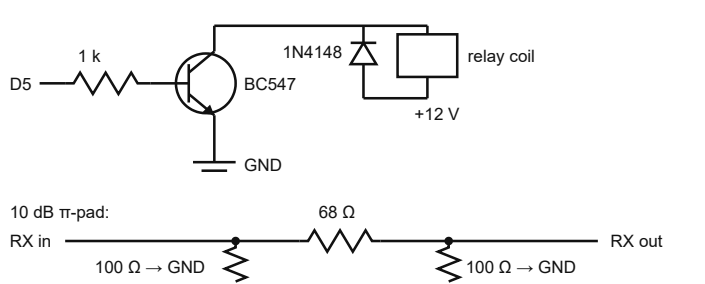
3. Buttons on A6 (resistor ladder)



Button	R to GND	ADC	Accept if
ATT on/off	0 Ω (direct)	0	< 90
MODE LSB/USB	2.2 k	184	90–255
AGC fast/slow	4.7 k	327	255–420
SPARE	10 k	512	420–770
none	—	1023	> 770

A6/A7 have **no internal pull-up** → the external 10 k is required. One button at a time. Resistors 1 % or 5 % both fine.

4. Attenuator relay (D5) — RX path only



- π-pad 100 / 68 / 100 Ω ≈ 10 dB (exact: 96 / 71 / 96 Ω). Relay bypasses or inserts it.
- **Only in the receive line after the antenna relay — never in the TX path.** Short leads / thin coax.
- 5 V relay: coil to +5 V instead of +12 V. Diode cathode (band) to + side.

5. AGC fast/slow (D13) — later

- 2N7000: gate via 100 Ω from D13, 100 k gate–GND, source GND, drain via R_{fast} (~150–330 k, to try) to the AGC hold node on PC-300D.
- Parts on PC-300D to be identified by value first (hold cap / 1 M discharge R).

6. MODE LSB/USB

Software only: BFO moved to the other side of the crystal filter (5.520.000 ↔ ~5.523.3 MHz, to be set) — LO calculation follows automatically.